DVLSI Assignment 6: 16-bit Brent–Kung Adder

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***Abstract*—This report presents the design, schematic, layout, and post-layout characterization of a 16-bit Brent–Kung adder using the standard cells developed in the previous assignment. The Brent–Kung architecture achieves logarithmic carry computation delay with reduced wiring complexity compared to Kogge–Stone. The design was implemented in Cadence Virtuoso using Metal 5 for power routing and up to Metal 5 for signal routing. Post-PEX simulations were used to estimate delay, power, and energy across process corners.**

***Index Terms*—Brent–Kung Adder, Tree Adder, VLSI Layout, Characterization, FO4, Power, Delay.**

I. Introduction

This work, done in collaboration with Pranjal Upadhyay (M.Tech, MVLSI), focuses on realizing a high-speed 16-bit adder using the **Brent–Kung topology**. The architecture provides a balance between the wiring complexity of Kogge–Stone and the latency of a Ripple-Carry Adder (RCA). It achieves a depth of $2\log_2 N - 1$ stages, where N is the number of bits.

All cells used were drawn from the mini standard cell library developed in Assignment 5, including inverters, NAND, NOR, and XOR gates. Each bit's **Propagate (P)** and **Generate (G)** signals were formed, which were then combined through a tree structure to compute carries efficiently.

II. Brent–Kung Architecture Overview

The Brent–Kung adder operates in two main phases:

- **Pre-computation:** Compute local propagate and generate pairs:
$$P_i = A_i \oplus B_i, \quad G_i = A_i \cdot B_i$$
- **Prefix Tree Computation:** Compute the prefix operations using:
$$(G_i, P_i) \otimes (G_j, P_j) = (G_i + P_i G_j, P_i P_j)$$
- **Sum Generation:**
$$S_i = P_i \oplus C_i$$

The prefix network was realized hierarchically:

- **Level 1:** 16 PG generation cells.
- **Level 2–5:** Hierarchical prefix operations forming a binary tree.
- **Level 6:** Sum and Cout computation.

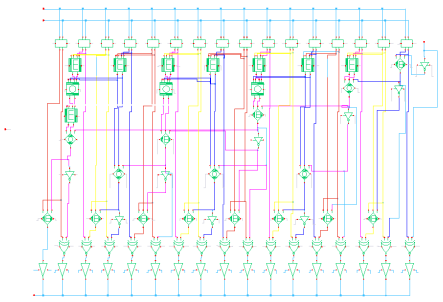


Fig. 1: Schematic of the 16-bit Brent–Kung Adder showing propagate/generate tree and carry computation.

III. Layout and Routing Strategy

Layout Hierarchy: Each functional block (PG cell, black/gray cell, and sum cell) was instantiated as a separate layout cell to preserve modularity and reduce error during LVS/DRC.

Power Routing:

- M5 used for global VDD and GND stripes (max width, 50 µm spacing).
- Tap-downs from M5 to M1 rails at regular intervals.

Routing Rules:

Layer	Preferred Direction
M1	Any
M2	Horizontal
M3	Vertical
M4	Horizontal
M5	Vertical

Pin Placement: All input and output pins were placed in Metal 5 at the top and bottom edges as per design rules.

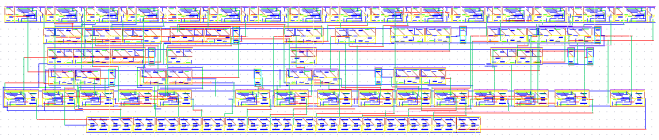


Fig. 2: Layout of the 16-bit Brent–Kung Adder (post-DRC clean).

IV. Characterization and Results

The design was simulated using extracted parasitic values (post-PEX). The output load was a 2 fF capacitor plus a minimum inverter to emulate realistic driving conditions.

TABLE I: Post-Layout Characterization Results

Corner	tpd (ps)	Power (μW)	Energy per cycle (fJ)	Area (μm^2)
Fast (1.1V, -40 °C)	630	592	592	218.3
Typical (1.0V, 27 °C)	1041	387	387	218.3
Slow (0.95V, 127 °C)	1125	288	288	218.3

V. Conclusion

A 16-bit Brent–Kung adder was successfully designed, laid out, and characterized using Cadence Virtuoso. Almost 300 percent increase was observed in delay post PEX.

References

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